Procedure:

Dissolve one tablespoon or 1 gram of soil from each soil sample in 100 ml of distilled water in separate beakers. Stir the solutions well and keep aside for half an hour to settle down the suspended particles. Filter off each solution separately in different test tubes. Dip a small piece of broad range pH paper on each of the solution. Match the colour of the pH paper with the colour scale given on the pH paper booklet. This gives an approximate pH.

Observation:

Record the pH of different soil samples in the observation table.

S. No.	Soil sample	pH Value
1		
2		
3		



Figure 21: Study of pH of different types of soil

Inference:

Thus the pH value of different soil samples required for plant growth can be determined.

Precautions:

1. Wash the glassware thoroughly and get it dried before the experiment.
2. Dry the pH papers before comparing the colour with the colour scale.
3. Match the colour carefully and determine pH accurately.

Exercise 22: Water holding capacity of garden soil and roadside soil

The maximum amount of water retained by soil per unit of its dry weight after the gravitational flow has ceased is called water holding capacity or field capacity of the soil. The water holding capacity varies in different type of soils and depends upon the types of soil particles and porocity of the soil. Sandy soils have poor water holding capacity then the loam and clay soils.

Aim:

To study the water holding capacity of garden soil and roadside soil.

Requirements:

Garden soil, roadside soil, measuring cylinders, funnels, filter papers, beakers, balance, etc.

Procedure:

Take two funnels and line them with filter paper. Lable them A and B. Place them on measuring cylinders. Take 100 gm dried sample each of the garden soil and roadside soil. Put the garden soil in funnel A and roadside soil in funnel B. Pour 100 ml of water in each funnel. Record the volume of filtered out water in the measuring cylinder when the dripping of water stops from the funnel.

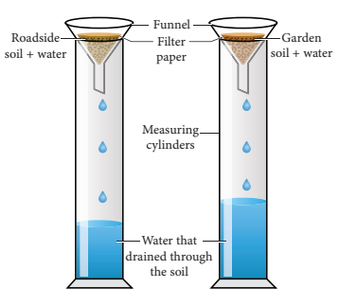


Figure 22: Water holding capacity of soil

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